

Strategic Product Placement Analysis System

1. INTRODUCTION

1.1 Project Overview

The Strategic Product Placement Analysis System is a B2B SaaS platform designed to revolutionize how physical retail spaces are optimized. By bridging the gap between raw point-of-sale (POS) data and physical store layouts, the system empowers retail merchandisers and store managers to make data-driven decisions without the need for expensive IoT hardware or smart-camera installations.

Physical retail environments inherently lack the granular, automated "click-tracking" analytics found in e-commerce. Currently, retail merchandisers struggle to translate raw, 2D POS spreadsheet data into optimal physical 3D display strategies. Mentally mapping thousands of rows of sales data into a physical space is cognitively difficult, leading to sub-optimal cross-merchandising, reliance on trial-and-error, and ultimately, lost revenue opportunities.

We are developing a hardware-free, AI-driven spatial mapping web platform. The system ingests standard 2D store floor plans (PDF/PNG) alongside historical POS CSV data. Utilizing a custom Spatial Engine, it maps sales density to physical aisles and generates an interactive D3.js spatial heatmap overlay. This allows users to easily identify high-traffic "hotspots" and underutilized "cold zones," predicting the best locations to cross merchandise premium products to disrupt shopper autopilot.

1.2 Purpose

The primary purpose of the Strategic Product Placement Analysis System is to democratize spatial retail analytics. By transforming flat, disconnected point-of-sale (POS) data into actionable, 3D spatial intelligence, the project aims to help retail merchandisers and store managers optimize in-store product placements without the need for prohibitive capital investments in smart-hardware.

- **Drive Revenue through Data-Backed Merchandising:** To eliminate "guesswork" and trial-and-error in retail layout planning. By predicting and highlighting high traffic zones, the system purposefully enables retailers to position premium and complementary products in exact locations that disrupt shopper "autopilot," thereby maximizing spontaneous impulse purchases and increasing average basket sizes.
- **Provide a Hardware-Free Analytics Alternative:** Existing spatial analytics solutions rely heavily on expensive, privacy-intrusive IoT sensors or smart cameras. A core purpose of this system is to provide a purely software-driven alternative. It leverages

existing, underutilized assets—standard 2D architectural floor plans and historical POS spreadsheet data—to deliver equivalent spatial insights at a fraction of the cost.

- **Bridge the Gap between Corporate and Store-Level Execution:** To create a cohesive feedback loop between corporate retail data analysts and local brick-and-mortar store managers. The system is designed to seamlessly translate complex mathematical AI predictions into simple, visually intuitive heatmap overlays and PDF strategy guides that non-technical staff can easily implement on the store floor.

2. IDEATION PHASE

2.1 Problem Statement

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Retail Store Manager	Increase product sales by placing products in the best locations.	Customers are not noticing many products, resulting in lower sales.	Product placement decisions are made without analyzing sales and promotional data.	Frustrated because sales and customer satisfaction are decreasing.
PS-2	Business Owner	Identify the best performing products and promotions.	It is difficult to identify the best performing products and promotions from raw sales data.	There is no interactive dashboard for data analysis.	Confused and unable to make timely, data driven business decisions.

2.2 Empathy Map Canvas

An empathy map is a visual framework used to understand the needs, behaviors, thoughts, and challenges of users. In this project, the empathy map helps identify the expectations and problems faced by retail store managers and business owners while making product placement decisions.

- **Think & Feel:** Wants to increase product sales. Wants better product visibility. Needs accurate business insights. Wants to make data-driven decisions.

- **See:** Low-performing product categories. Crowded shelves and poor product placement. Promotional offers in competitor stores. Sales dashboards and reports.
- **Hear:** Customer feedback about products. Suggestions from the sales team. Market trends and competitor strategies. Sales performance reports.
- **Say & Do:** Analyze sales reports regularly. Compare product performance. Rearrange product placement. Monitor promotional campaigns.
- **Pain:** Low sales due to poor product placement. Difficult to identify best-selling products. Manual analysis is time-consuming. Poor customer engagement.
- **Gain:** Increased sales and revenue. Better customer shopping experience. Faster and data-driven decision making. Optimized product placement strategy.

2.3 Brainstorming

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon.

- **Cluster 1: Cross-Merchandising Synergy:** Place our premium beverage products directly in the fresh bakery section. Position our complementary snack items on the same endcap as major beverage brands. Use clip-strips to hang our products next to high-volume seasonal items.
- **Cluster 2: High-Traffic Checkout Disruptions:** Secure eye-level shelving specifically at self-checkout kiosks. Install branded mini-coolers at the end of every traditional register lane. Create floor decals leading from the store entrance directly to the checkout impulse racks.
- **Cluster 3: Experiential & Interactive Zones:** Set up weekend sampling stations staffed by brand ambassadors in the main center aisles. Install interactive QR-code shelf-talkers that offer an instant digital coupon. Use digital endcap screens that play our product commercials when motion is detected.

Top Priorities: Cross-merchandising our premium beverages in the fresh bakery section. Securing eye-level shelving at self-checkout kiosks. Positioning complementary snack items on endcaps with major brands.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Phase	Steps	Interactions	Goals & Motivations
Entice	Enters the retail store, grabs a shopping cart, and notices entrance promotional signage or seasonal displays.	Things: Store entrance, carts, digital entrance screens, initial promotional flyers.	"Help me get my bearings and see if there are any immediate good deals today."
Enter	Navigates the main aisles, heading towards their primary target categories (e.g., fresh bakery or main grocery).	Places: High-traffic perimeter aisles, fresh food sections. Things: Aisle markers.	"Help me find my routine items quickly without getting stuck in crowded aisles."
Engage	Spots the cross merchandised premium product line on an endcap or interactive shelf-talker. Stops to inspect.	Things: QR-code shelf talkers, complementary snack cross-merchandising displays, product packaging.	"Help me understand why this premium product is worth the extra cost over my usual brand."
Exit	Proceeds to the self checkout kiosk. Notices secondary placement of the product. Completes purchase.	Places: Self-checkout zone. Things: Eye-level shelving at the kiosk, POS system, digital payment.	"Help me check out fast without hassle, but maybe grab a last minute treat."

3.2 Solution Requirement

Functional Requirements:

- FR-1 User Management & Authentication: Registration through Corporate Email, Role-based Login, Password Recovery.
- FR-2 Data Ingestion Module: Upload POS Sales Data, Upload Store Floor Plan Images.
- FR-3 Spatial Mapping & Analysis Engine: Generate digital coordinate grid over uploaded floor plan, Assign product sales metrics, Run predictive algorithm.
- FR-4 Heatmap Visualization: Render interactive D3.js heatmap overlay, Toggle visibility layers.
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FR-5 Reporting & Export: Generate PDF Strategy Report, Export raw placement coordinate data.

Non-functional Requirements:

- NFR-1 Usability: Intuitive UI/UX requiring less than 1 hour of training.
- NFR-2 Security: Hashed passwords (bcrypt), Encrypted data (AES-256), HTTPS (TLS 1.2+).
- NFR-3 Reliability: 99.9% uptime during retail business hours.
- NFR-4 Performance: Capable of parsing 50MB CSV files and rendering heatmap within 3 seconds.
- NFR-5 Availability: Accessible 24/7 across major modern browsers.
- NFR-6 Scalability: Support concurrent spatial processing for up to 100 distinct retail floor plans.

3.3 Data Flow Diagram

1. Data Ingestion: The Retail Analyst uploads raw store floor plan files (PDF/PNG) and historical POS sales data (CSV) via the web interface.
2. Processing & Mapping: The system's Spatial Engine parses the floor plan to generate a digital grid, mapping the POS data to specific physical aisle coordinates.
3. Enrichment: Mapped data is passed to the AI Analysis Module which calculates high traffic zones and predicts impulse-buy hotspots.
4. Visualization: The enriched dataset is returned to the client and visualized as an interactive Heatmap Dashboard overlaying the original floor plan using the D3.js library.
5. Export: The user requests a strategy report; the system compiles the visual data and metrics into a PDF payload and delivers it to the user's device.

3.4 Technology Stack

- **User Interface:** React.js, HTML5, CSS3, D3.js (for Heatmaps)
- **Application Logic:** Node.js (Express) or Python (FastAPI)
- **Spatial Engine:** Python, OpenCV, Pandas
- **Database:** PostgreSQL (Relational), MongoDB (NoSQL)
- **File Storage:** AWS S3 / Cloud Object Storage

- **Machine Learning:** Python, Scikit-Learn, TensorFlow
- **Infrastructure:** Docker, Kubernetes, AWS/GCP

4. PROJECT DESIGN

4.1 Problem Solution Fit

- **Customer Segments:** Retail Merchandisers, Retail Data Analysts, Store Managers.
- **Jobs-to-be-Done:** Identify Hotspots, Disrupt Autopilot, Data Synthesis, Prove ROI.
- **Problem Root Cause:** Physical retail environments inherently lack the granular, automated "click-tracking" analytics found in e-commerce. Mentally mapping thousands of rows of 2D sales data into a physical 3D space is cognitively impossible for humans at scale.
- **Your Solution:** Hardware-Free Spatial Engine, AI Heatmapping, Predictive Placements, Exportable Strategy.

4.2 Proposed Solution

A hardware-free, AI-driven spatial mapping web platform. It ingests standard store floor plans (PDF/PNG) and POS sales data (CSV) to generate interactive heatmaps. The system predicts impulse-buy hotspots and recommends optimal aisle placements for secondary displays.

Unlike existing solutions that require massive capital expenditure and logistical hurdles (e.g., installing IoT shelf sensors or smart-tracking cameras), this system leverages existing historical POS data and standard architectural floor plans to provide a purely software based, low-cost spatial analytics engine.

B2B SaaS (Software as a Service) subscription model. Pricing is tiered based on the number of retail locations analyzed or the volume of POS data processed monthly. Includes potential one-off onboarding fees for initial floor-plan digitization.

4.3 Solution Architecture

This architecture bridges the business need for automated retail spatial analytics with a robust, scalable cloud infrastructure. It defines how data (floor plans and POS sales) is ingested, processed by a backend spatial engine, and visualized as actionable intelligence for retail analysts and managers.

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1. Input: Retail Analyst uploads Floor Plan (PDF/PNG) & POS Sales Data (CSV).
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2. Ingestion: Data is routed via API Gateway to Backend Microservices.
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3. Processing: Spatial Engine maps POS sales coordinates onto the Floor Plan grid.
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4. Intelligence: ML Models calculate high-traffic zones; enriched data stored in MongoDB.
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5. Visualization: Frontend UI pulls enriched data and renders Heatmap overlay using D3.js.
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6. Security: All processed data encrypted via Key Management Service (KMS).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epic	Task	Points
Sprint 1	Registration & Login	As an analyst, I can log into the web application securely using corporate credentials.	3
Sprint 1	Data Ingestion	As an analyst, I can upload CSV files containing sales data and store floor plan images.	5
Sprint 2	Spatial Mapping	As the system, I will map POS data coordinates to the uploaded digital floor plan grid.	8
Sprint 2	Heatmap Visualization	As an analyst, I can toggle an interactive heatmap overlay to identify trafficked aisles.	5
Sprint 3	Dashboard	As a store manager, I can view finalized product placement recommendations on mobile.	5
Sprint 3	Reporting	As an analyst, I can export the placement strategy as a PDF report.	3

Velocity Calculation: Total Story Points = 10 + 13 + 7 = 30. Number of Sprints = 3. Velocity = 10 Story Points per Sprint.

5.2 PROJECT DEVELOPMENT PHASE 1

Data Preprocessing Steps:

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Step 1: Dataset Collection: The dataset used in this project, "Impact of Product Positioning on Sales," was downloaded from Kaggle in CSV format. It contains retail sales data related to product positioning, product pricing, competitor pricing, promotional activities, customer demographics, seasonal trends, foot traffic, and sales volume.

- **Step 2: Opening the Dataset:** The downloaded CSV file was opened in Microsoft Excel to understand the structure of the dataset and verify the available fields before importing it into Tableau Public.

- **Step 3: Importing Dataset into Tableau Public:** The CSV file was imported into Tableau Public using the Text File connection. Tableau successfully loaded the dataset containing 1000 records.

- **Step 4: Renaming the Field:** To improve readability and maintain consistency throughout the dashboard, the field "Competitor's Price" was renamed to "Competitor Price".

- **Step 5: Verifying Data Types:** After importing the dataset, the data types of the required fields were verified and modified where necessary.

- **Step 6: Creating Calculated Fields:** The required calculated fields were created in Tableau Public to support dashboard development and improve data visualization. •

- **Step 7: Dataset Ready for Visualization:** After importing the dataset, renaming the required field, verifying the data types, and creating the calculated fields, the dataset was ready for creating worksheets, dashboards, and Tableau Stories.

5.3 PROJECT DEVELOPMENT PHASE 2

- **1. Tableau Dashboard:** The dashboard combines multiple visualizations into a single interactive interface for analyzing product placement, pricing, promotions, customer demographics, seasonal trends, and sales performance.

- **2. Tableau Story:** The Tableau Story presents the complete business analysis step by step using interactive story points, making it easier to understand key business insights.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Parameter	Values / Results
Data Rendered	File Name: Product Positioning.csv. File Type: CSV. Total Records: 1000. Total Columns: 10.

Data Preprocessing	Missing Values: 0. Duplicate Records: 0. Data Types Verified: Integer, Float and String. Splitting/Merging: Not Performed.
Utilization of Filters	Filters used across different worksheets: Promotion, Foot Traffic, Consumer Demographics, Product Position, Product Category, and Measure Names.
Calculation fields Used	Calculated Field Name: Zero Syntax: 0. Purpose: Used as a constant calculated field for creating the donut chart visualization in the Tableau dashboard.
Dashboard design	Number of Dashboards: 1. Number of Visualizations/Graphs: 8. Interactive dashboard developed with filters and legends for business analysis.
Story Design	Number of Stories: 1. Number of Visualizations/Graphs: 8. Tableau Story created to present business insights in a logical sequence.

7. RESULTS

7.1 Output Screenshots

Note: Please refer to Annexure A for these screenshots.

8. ADVANTAGES & DISADVANTAGES

Advantages (Strengths):

- **Cost-Effective & Hardware-Free:** By completely bypassing the need for IoT shelf sensors, smart-cameras, or physical beacons, the system drastically reduces upfront capital expenditure (CapEx) for retail chains.
- **High Scalability:** As a cloud-native SaaS platform, it can be instantly deployed to analyze hundreds of stores nationwide without the logistical hurdles of physical hardware installation.
- **Privacy Compliant:** Because the system maps aggregate POS sales data rather than tracking individual shoppers via cameras or Wi-Fi, it inherently avoids strict GDPR and consumer privacy concerns.
- **Leverages Existing Assets:** Maximizes the value of data retailers already collect (historical POS spreadsheets) and assets they already possess (standard 2D architectural floor plans).
- **Actionable & Accessible:** Demystifies complex spatial data for non-technical users,

generating simple PDF layout guides that store managers can implement immediately to drive ROI.

Disadvantages (Limitations):

- **Lack of Real-Time Foot Traffic Data:** Because it relies on historical POS sales data, the system cannot track live foot traffic or account for "window shoppers" who walk through an aisle but do not make a purchase.
- **Dependency on Data Quality:** The accuracy of the AI heatmaps is strictly bound by the quality of the uploaded POS data and how accurately the products are mapped to the digital floor plan. Discrepancies lead to flawed recommendations.
- **Floor Plan Upkeep:** Requires accurate, digitized 2D floor plans. If a physical store has undergone layout changes that are not reflected in the uploaded PDF/PNG, the spatial mapping will be misaligned.
- **Indirect Routing Assumptions:** The system identifies "hotspots" based on where items were sold, but it cannot map the exact physical walking path a customer took to get to that aisle, missing nuanced shopper navigation behaviors.

9. CONCLUSION

The Strategic Product Placement Analysis System successfully bridges a critical gap in physical retail analytics by providing a cost-effective, hardware-free alternative to traditional foot-traffic tracking. By leveraging existing datasets rather than forcing retailers into massive capital expenditures for IoT sensors or smart-cameras, this project democratizes advanced retail analytics. The system automates the cognitively heavy task of translating flat spreadsheet rows into interactive, 3D/2D spatial heatmaps, drastically reducing the reliance on "trial and error" merchandising. The solution addresses the direct needs of corporate merchandisers, data analysts, and local store managers alike. Corporate teams can now mathematically justify premium shelf-space negotiations and cross-merchandising strategies using concrete visual evidence. Simultaneously, local store managers receive clear, exportable PDF guidelines that eliminate the guesswork of where to set up secondary impulse displays to disrupt "shopper autopilot." In conclusion, the Strategic Product Placement Analysis System stands as a highly scalable, privacy compliant, and immediately actionable tool that brings e-commerce level precision to physical brick-and-mortar retail environments.

10. FUTURE SCOPE

- **Real-Time ERP & Inventory Integration:** Future iterations will connect directly with

enterprise retail inventory management systems (e.g., SAP, Oracle). This will transition the platform from relying on static CSV uploads to pulling live data streams.

- **Predictive & Seasonal Scenario Modeling (A/B Simulator):** By incorporating external data sets such as local weather patterns, seasonal trends, and regional events, the AI engine will evolve from identifying past hotspots to predicting future ones.
- **Augmented Reality (AR) & 3D Visualizations:** To further bridge the gap between flat data and physical reality, the 2D D3.js heatmaps will be upgraded to support 3D WebGL rendering. Using Augmented Reality (AR) via mobile devices, store managers will be able to hold up their tablets in-store and see the digital heatmaps superimposed directly over the physical retail aisles in real-time.
- **Omnichannel Shopper Behavior Sync:** The ultimate goal is to create a seamless omnichannel layout strategy. By linking online e-commerce click-data (e.g., products frequently bought together online) with the physical store's spatial engine, the system will automatically recommend physical cross-merchandising setups that mirror successful digital basket trends.

11. APPENDIX

- **Dataset Link:**
<https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>
- **Tableau Public Dashboard Link:**
https://public.tableau.com/views/StrategicProductPlacementAnalysis1_17836935205570/Dashboard1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- **Tableau Public Story Link:**
https://public.tableau.com/views/StrategicProductPlacementAnalysis1_17836935205570/Story1?:language=en-GB&:sid=&:display_count=n&:origin=viz_share_link

Annexure A

Project Output Screenshots

1. Strategic Product Placement Analysis Dashboard

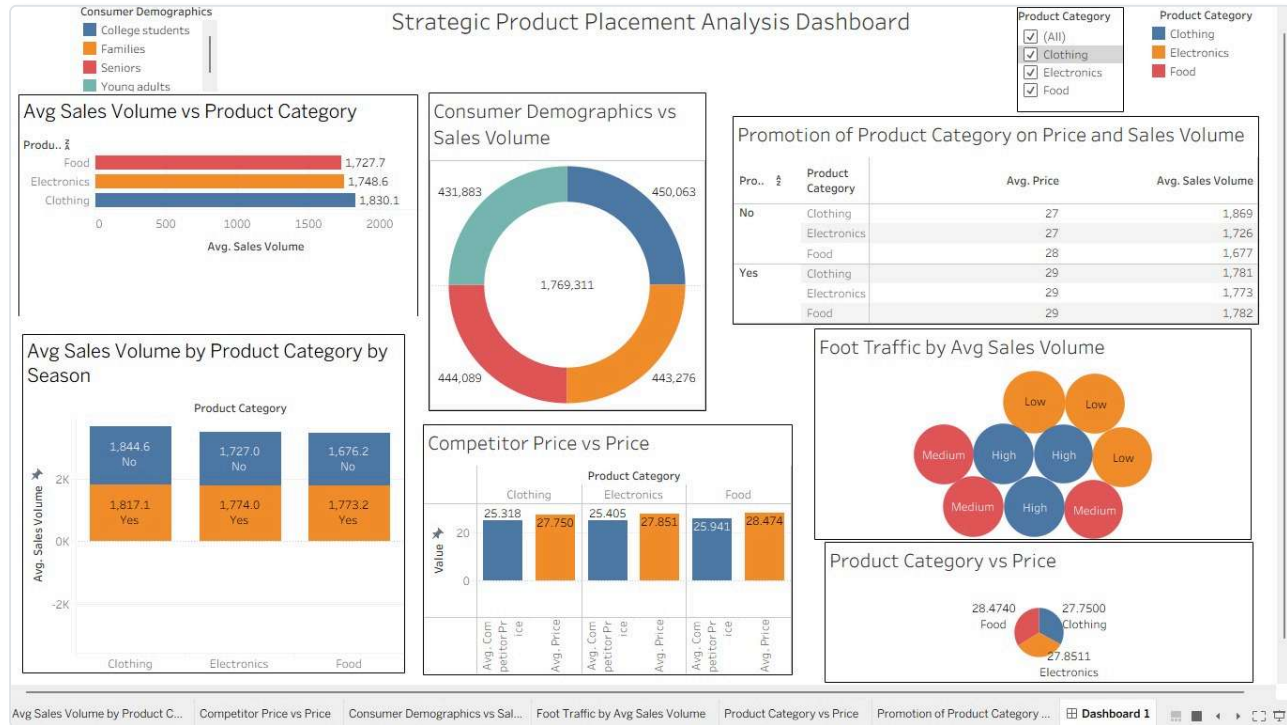


Figure 3: A comprehensive view of the Tableau dashboard, featuring various metrics such as sales volume by category, consumer demographics, and foot traffic.

2. Average Sales Volume by Product Category

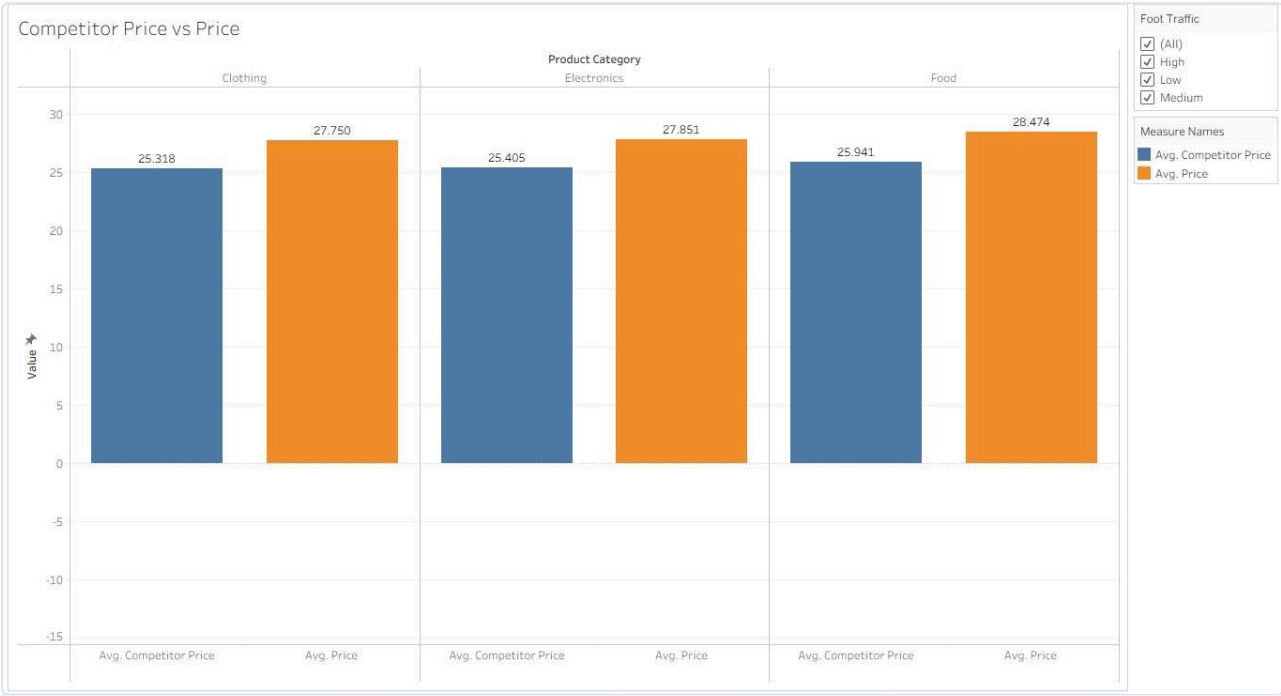


Figure 4: A horizontal bar chart displaying the average sales volume for Food, Electronics, and Clothing categories.

3. Average Sales Volume by Product Category by Season

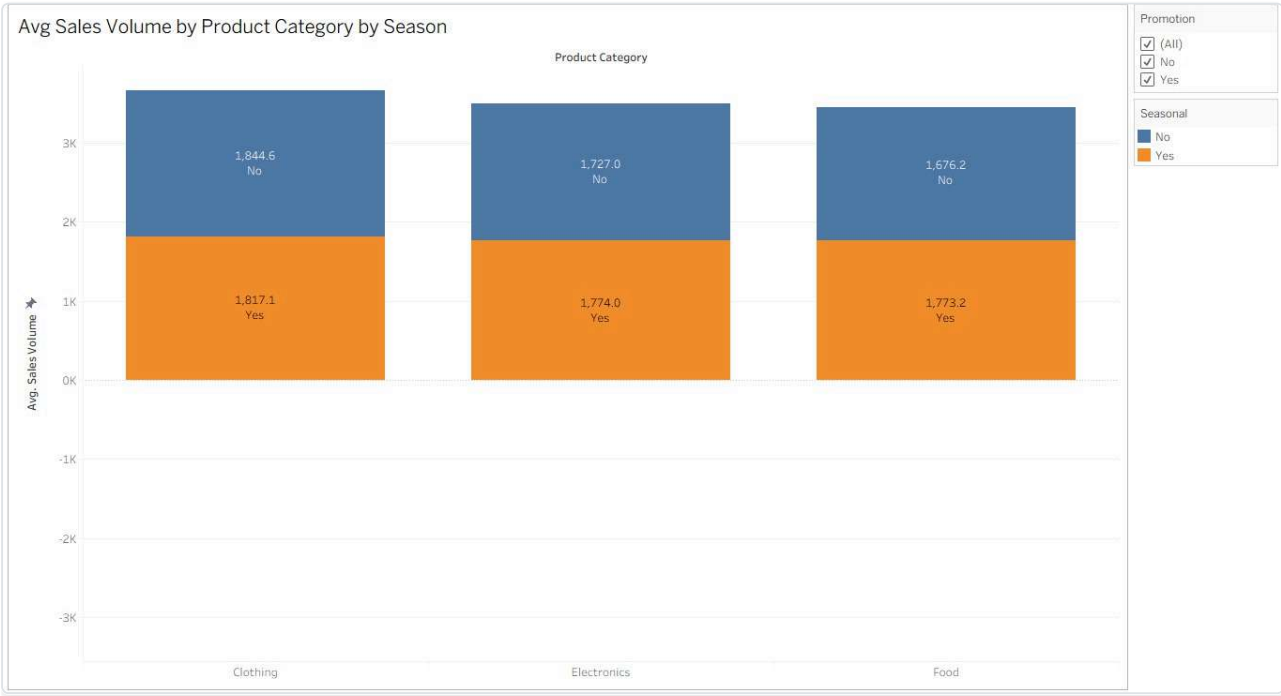


Figure 5: A stacked bar chart breaking down average sales volume by product category and seasonal influence.

4. Consumer Demographics vs Sales Volume



Figure 6: A donut chart illustrating the distribution of sales volume across different consumer demographic segments.

5. Competitor Price vs Price

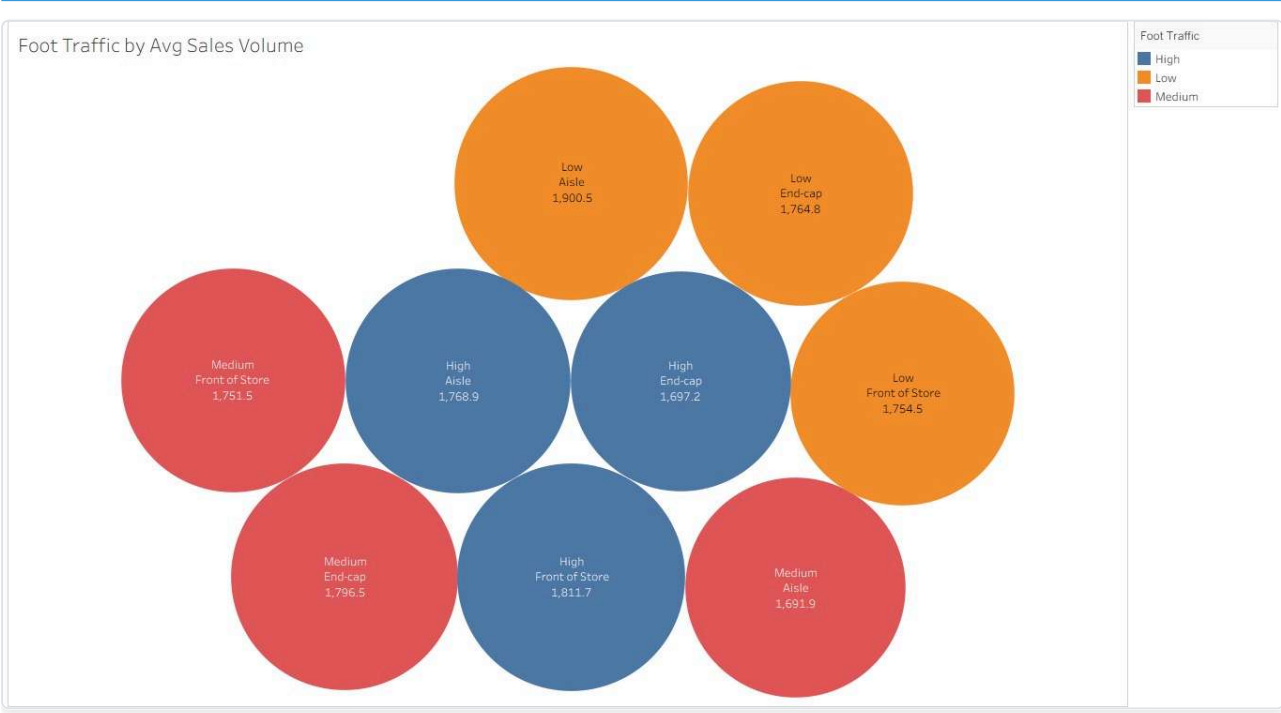


Figure 7: A side-by-side bar chart comparing the average competitor price against the average internal price across different product categories.

6. Foot Traffic by Average Sales Volume

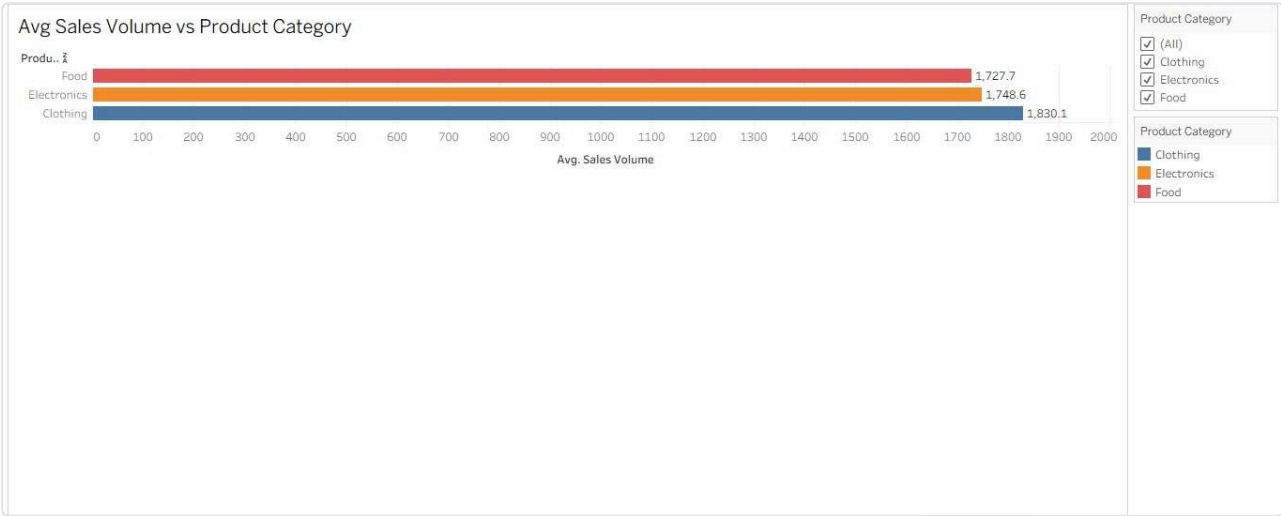


Figure 8: A bubble chart visualizing the relationship between foot traffic intensity (High, Medium, Low) and average sales volume.

7. Promotion of Product Category on Price and Sales Volume

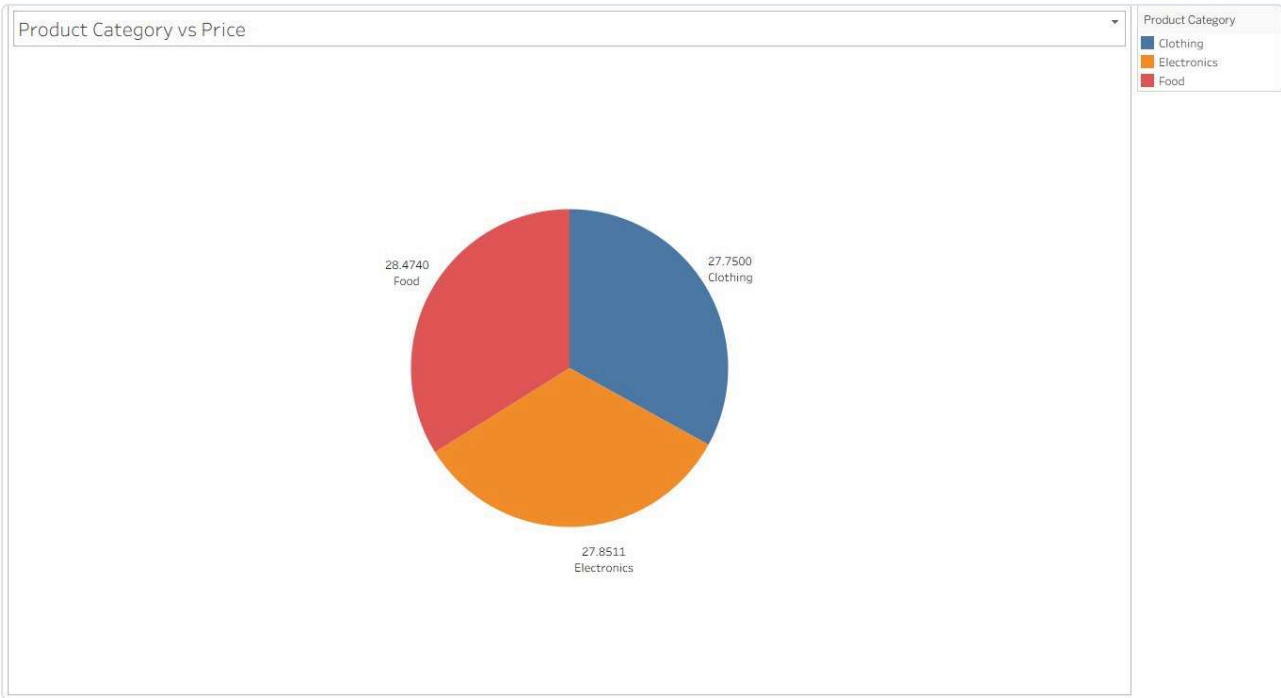


Figure 9: A tabular view analyzing the impact of promotions on the average price and sales volume for various product categories.

8. Average Sales Volume by Product Category by Product Position

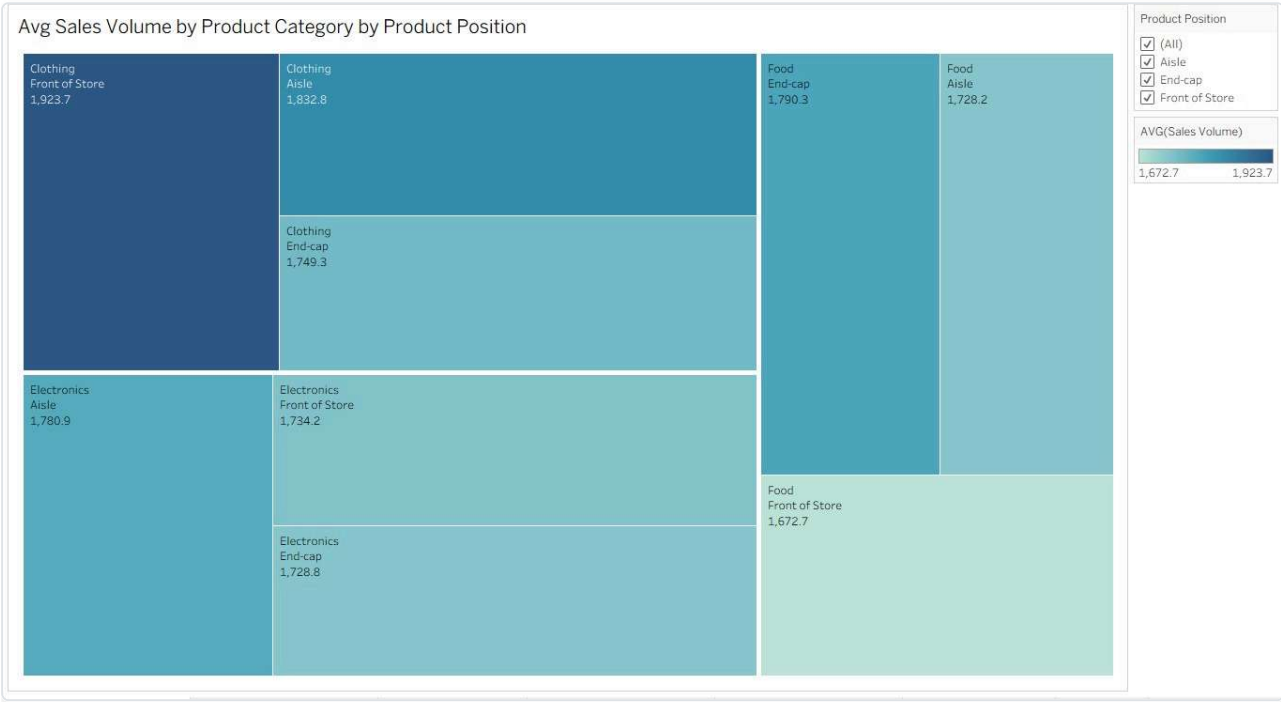


Figure 10: A treemap visualization showing how average sales volume varies based on the product's physical position in the store (Aisle, End-cap, Front of Store).

9. Product Category vs Price

Promotion of Product Category on Price and Sales Volume

Pro..	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

Figure 11: A pie chart displaying the average price distribution among Clothing, Electronics, and Food categories.